

# Dinomaly & AnomalyDINO

Two recent anomaly-detection models, both built on the same frozen DINOv2 backbone. This note covers how each one was implemented here and what it scored, next to the pixel-difference detector the pipeline already uses. One of them needs a clean reference photo and no training. The other needs no reference, but a model trained for each camera.

## Findings

1. **AnomalyDINO draws the best boxes.** It covers 58/73 objects properly against 42/73 for the current detector. Almost all of the gap is tram 3333<sup>\*</sup>, where the current detector manages 5/26 and AnomalyDINO 19/26. It needs no training.
2. **Dinomaly draws weaker boxes (48/73) but removes bad ones well.** Used to filter AnomalyDINO's output it keeps the same 58/73 while cutting the regions sent on from 1228 to 654, so the VLM has 47 % fewer crops to judge.
3. **End to end, the number of objects reported does not change.** It stays at 55 of 73 with every option. What improves is the share of anomalous frames caught, 0.871 to 0.968, and the share of reported regions that are real, 0.694 to 0.896.
4. **Dinomaly needs one trained model per camera**, 59 MB each, and empty footage of that camera to train on. That footage is the binding constraint, not the compute: training takes 1 min 42 per camera.

## METHOD

### The two models

Both models replace the same step. Before any VLM is asked *what* is in a region, something has to decide *where* to look. Both run the same frozen DINOv2 ViT-S/14-reg encoder, which turns an image into a grid of patch descriptors. What differs is what they compare those patches to, and that one choice decides how each of them can be used in practice.

#### REFERENCE-BASED

### AnomalyDINO

WACV 2025. A patch is flagged when it no longer looks like the patch that sat in the same spot on the clean reference photo.

|          |                                      |
|----------|--------------------------------------|
| TRAINING | none                                 |
| NEEDS    | one clean reference frame per camera |
| STATE    | none per camera                      |
| COST     | ≈2 s per frame                       |

#### REFERENCE-FREE

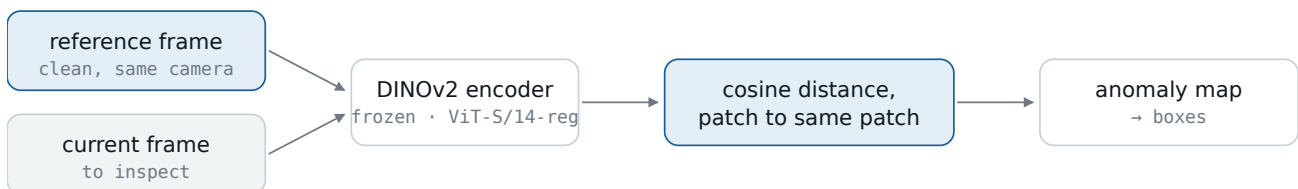
### Dinomaly

CVPR 2025. A patch is flagged when a decoder trained on this camera's empty frames fails to rebuild it.

|          |                                   |
|----------|-----------------------------------|
| TRAINING | 1 min 42 per camera (RTX 3080 Ti) |
| NEEDS    | empty footage of that camera      |
| STATE    | 59 MB per camera                  |
| COST     | ≈3 s per frame                    |

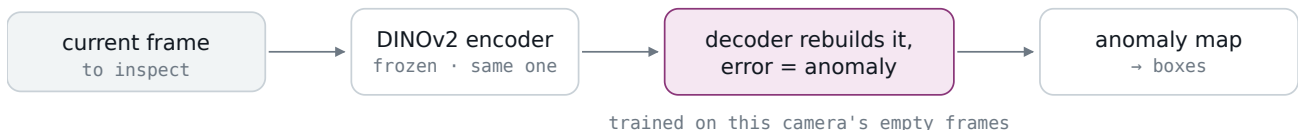
### AnomalyDINO

needs a clean reference frame · no training



### Dinomaly

no reference at inference · one trained model per camera



Everything after the anomaly map is shared with the current detector: the same region finder, the same person filter, the same merging step. That way a comparison between the two models only measures the change signal itself.

#### METHOD

## AnomalyDINO: implementation

- 01 Encode both frames.** The reference photo and the current image are resized to 1120 px wide and passed through the frozen encoder. Each one comes out as an 80×45 grid of patch descriptors. On a 1920×1080 frame a patch covers 24 px, on a 1280×720 frame 16 px. That grid is also the smallest box either model can draw.
- 02 Compare each patch to the same spot, not to a pool of patches.** This is where our version departs from the paper. The paper collects all the normal patches into one memory bank, because the objects it works on are centred but not aligned. Our cameras are bolted to the tram, so we can be stricter: a patch is only compared to reference patches within one patch of its own position. It makes for a much tighter test. A seat cushion gets compared to *that* seat, not to any seat in the picture.

**03 Use two thresholds, not one.** The distance map is normalised by its own median and spread, then a patch is flagged only if it clears both a relative threshold (4 times the map's typical spread, measured in a way that outliers cannot inflate) and an absolute one (distance 0.08).

Both are needed. The spread of the map varies by a factor of five across our cases, so a single relative threshold cuts at 0.043 on one frame and 0.225 on another. On two nearly identical frames the spread collapses and the relative threshold alone invents 80 regions or more. The absolute floor stops that, and the relative one handles frames shot in different light.

## METHOD

### Dinomaly: implementation and training data

**01 Freeze the encoder, train only a decoder.** 15.4 M trainable parameters sitting on top of the frozen DINOv2. We use eight of its twelve middle blocks, split into two groups. The last blocks are too specialised and the first ones too generic to be useful here.

**02 Use linear attention, and not for speed.** Standard attention is sharp enough that a decoder token can simply copy the encoder token it is meant to rebuild, anomaly included, which would defeat the whole idea. A smoother kernel cannot pick out a single token, so an unfamiliar patch has to be rebuilt from its surroundings. It fails, and that failure is the signal we detect.

**03 Train one model per camera, on empty frames.** 1282 frames over 8 models, 40 epochs each. A model of what *this* view normally looks like says nothing about another view, so covering the fleet's 26 cameras would mean 26 models.

**04 Score by how badly the rebuild fails.** At inference the frame is encoded, the decoder rebuilds it, and the gap between the two is the anomaly map. No reference photo is used at any point.

| CAMERA     | FRAMES | SESSIONS | FROM                         | TRAINING LOSS |
|------------|--------|----------|------------------------------|---------------|
| 1760-cam04 | 192    | 1        | single run (192)             | 0.83 → 0.030  |
| 1760-cam06 | 194    | 1        | single run (194)             | 0.82 → 0.030  |
| 1760-cam13 | 194    | 1        | single run (194)             | 0.83 → 0.030  |
| 3333-cam52 | 90     | 1        | 08-55-37 (90)                | 0.92 → 0.043  |
| 3333-cam53 | 173    | 2        | 08-55-37 (90), 08-59-54 (83) | 0.83 → 0.031  |
| 3333-cam54 | 151    | 2        | 08-55-37 (90), 08-59-54 (61) | 0.86 → 0.033  |
| 3333-cam55 | 153    | 2        | 08-55-37 (91), 08-59-54 (62) | 0.85 → 0.032  |
| tram_1762  | 135    | 2        | v1 (68), v3 (67)             | 0.88 → 0.031  |

These numbers are read back out of the checkpoint files, which each store the list of images they were trained on. Every image is an empty tram, taken one every 2 seconds from the source video, then filtered twice. Anything shot within 15 seconds of a test image is dropped, so no model is ever scored on what it learned from. A person detector then removes any image with someone in it, which took out 18 % of the 3333 candidates. One 3333 recording was excluded by hand: it has two passengers *and* a bag on the floor, and the detector would have kept the frames where the bag sits alone.

#### The training data lacks variety, not quantity

The three 1760 models have around 190 images each, and all of them come from a single ten-minute run. Same hour, same light, same weather. Pulling five hundred more out of that video would just give five hundred nearly identical pictures. These models have never seen their own camera on another day, or at night.

RESULTS

Box quality

An object only counts here if one region covers it properly. Merely overlapping it is not enough, because a box spanning the whole image overlaps everything while telling the VLM nothing about where to look. No VLM was run for this table.

**On its own, AnomalyDINO is the best of the three**, with 58/73 against 42/73 for the pixel diff. The gap is not spread evenly. Almost all of it comes from tram 3333, where the pixel diff manages 5/26 and AnomalyDINO 19/26. On those cameras the pixel diff's worst box covers 99 % of the image. AnomalyDINO's worst is about ten times smaller.

**Dinomaly is weaker at drawing boxes** (48/73), but it is good at removing bad ones. Used to filter AnomalyDINO's output it keeps the same 58/73 while cutting the regions from 1228 to 654, so the VLM has 47 % fewer crops to look at. The two models make different mistakes, because one compares the image to a photo and the other to a model of the scene.

|                                                                                                   |                                                                 |                                                                                                        |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>58/73</b><br>objects boxed properly by AnomalyDINO, against 42/73 for the pixel diff           | <b>19/26</b><br>on tram 3333, where the pixel diff manages 5/26 | <b>-47 %</b><br>fewer crops sent to the VLM when Dinomaly filters AnomalyDINO, with no loss of quality |
| <b>1 min 42</b><br>to train one Dinomaly model on the RTX 3080 Ti, down from 56 min on the laptop |                                                                 |                                                                                                        |

| CONFIGURATION                                                          | MODEL USED           | BOXED PROPERLY | ON 3333 | REGIONS |
|------------------------------------------------------------------------|----------------------|----------------|---------|---------|
| <b>Pixel diff</b><br>the current detector, no neural model             |                      | 42/73          | 5/26    | 1192    |
| <b>AnomalyDINO alone</b><br>the model draws the boxes                  | ANOMALYDINO          | 58/73          | 19/26   | 1228    |
| <b>Pixel diff + AnomalyDINO filter</b><br>the model only deletes boxes | ANOMALYDINO          | 42/73          | 5/26    | 727     |
| <b>Dinomaly alone</b><br>the model draws the boxes                     | DINOMALY             | 48/73          | 19/26   | 813     |
| <b>Pixel diff + Dinomaly filter</b><br>the model only deletes boxes    | DINOMALY             | 42/73          | 5/26    | 699     |
| <b>AnomalyDINO + Dinomaly filter</b><br>one draws, the other deletes   | ANOMALYDINO DINOMALY | 58/73          | 19/26   | 654     |

All 68 cases. The “Regions” column is what gets sent on to the VLM. At roughly 15 seconds of CPU per region, it is what dominates the running time of a job. The shaded row is the current detector.

RESULTS

End to end, with the VLM

The table above counts boxes. This one puts the same 68 cases through the VLM that decides whether each box really contains an anomaly. It tells us what the better boxes were actually worth.

**The number of objects finally reported does not change.** It is 55/73 with every option. The four extra instances AnomalyDINO boxes are rejected by the judge anyway. What does change is the rest. The share of anomalous frames caught goes from 0.871 to 0.968, and the share of reported regions that are real goes from 0.694 to 0.896. In plain terms, wrong regions drop from 30 to 7. None of the options ever raises a false alarm on a clean frame.

| CONFIGURATION                          | REGIONS JUDGED | FRAMES CAUGHT | NO FALSE ALARM | OBJECTS | PRECISION |
|----------------------------------------|----------------|---------------|----------------|---------|-----------|
| <b>Pixel diff</b>                      | 1192           | 0.871         | 1.000          | 55/73   | 0.694     |
| <b>Pixel diff + AnomalyDINO filter</b> | 727            | 0.871         | 1.000          | 55/73   | 0.773     |
| <b>AnomalyDINO alone</b>               | 1228           | 0.968         | 1.000          | 55/73   | 0.896     |
| <b>AnomalyDINO + Dinomaly filter</b>   | 654            | 0.968         | 1.000          | 55/73   | 0.896     |

Judge: GLM-4.6V-Flash-9B with the conservative prompt, the pair every published number in this project uses.

## Limits

- **The three 1760 models only ever saw one session.** Each of those cameras has a single ten-minute run, and the test images come from that same run. Training images are kept 15 seconds away from any test image, but they still share its light. So nothing here tells us whether a 1760 model still works on another day.
- **The 3333 clean frames are not independent either.** Of the nine recordings, four contain the staged anomalies, one is the reference and one is used as a clean test case. The three that are unlabelled last 12, 8 and 10 seconds, which is far too short to train on. There is simply no clean pool left, so training uses the two long clean recordings minus the images held out. The 14 anomalous 3333 cases are genuinely from sessions the model never saw. The 7 clean ones are not, so the false-alarm rate on 3333 looks better than it should.
- **3333-cam52 is the weakest model of the eight.** It has only 90 images, all from one recording, and it ends training with the highest error.
- **On an unknown camera, Dinomaly goes quiet rather than wrong.** We showed it a scene it had never been trained on, and it filtered out zero regions instead of filtering out everything. In other words it falls back to doing nothing, not to blocking detections. An untrained camera costs us efficiency, not missed objects.
- **Box quality uses a 0.3 overlap threshold,** measured against boxes that were drawn by hand on a 100 px grid, fairly generously. It tells a usable box apart from a blob. It is not a precise measure of outline quality.

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ARSI · tram anomaly detection · region proposal

Pipeline: `benchmark/eval_localization.py` → `tools/collect_e2e.py` → `tools/dump_training_sets.py`

Raw data: `docs/dino_models/metrics.json`, `end_to_end.json`, `training_sets.json`

\* 3333 is a placeholder, not the tram's real fleet number - the vehicle number of the 2026-08-11 capture is unknown. It was called 39T before, but 39T is the Škoda type, which tram 1760 shares.